

ENGINEERING RELEASE GUIDE

Anticipy Hardware Handoff

Custom PCB EVT R0B

One controlled package for electrical, mechanical, firmware, manufacturing and final assembly

Release ANT-PROD-R0B-EVT-HANDOFF-2026-08-29

Prepared August 29, 2026

FABRICATION BLOCKED

The source is editable and useful, but 176 PCB connections remain unrouted. The battery, final enclosure and production firmware are also not released.

Release: ANT-PROD-ROB-EVT-HANDOFF-2026-08-29

1. What this package is

This package gives a senior electrical engineer, mechanical designer, firmware engineer, PCBA factory, and final assembler one controlled source of truth for turning the Anticipy custom-PCB concept into working EVT units.

It includes the editable KiCad project, custom footprints, draft BOM, pin map, verified board outline, provisional board-only STEP/DXF, engineering scope, manufacturing RFQs, firmware requirements, assembly plan, QA gates, and ready-to-send messages.

It does not pretend unfinished items are finished. The board is not fabrication-ready, the battery is not selected, and the final enclosure CAD and firmware binary do not exist yet.

2. Plain-English definitions

- **PCB:** the bare green circuit board.
- **BOM:** bill of materials, meaning the shopping list of electronic parts.
- **PCBA:** the PCB after the electronic parts have been soldered onto it.
- **Schematic:** the electrical map showing what connects to what.
- **PCB layout:** the physical placement and copper traces on the board.
- **Gerbers:** factory images of every PCB layer.
- **Drill files:** instructions for every plated hole, slot, and non-plated cut.
- **CPL or pick-and-place file:** the X/Y position, rotation, and side for every machine-placed part.
- **STEP:** a 3D CAD model used for collision and enclosure design.
- **DXF:** a 2D cutting profile, often used for sheet metal or outline transfer.
- **EVT:** engineering validation test, the first hardware revision used to prove the design works.

The PCB is not the BOM. The BOM is not the enclosure. The PCBA factory does not automatically create a finished pendant.

3. Frozen product intent

Anticipy is a camera-free wearable Bluetooth audio pendant. The device is intended to record ambient audio, store audio when offline, transfer live and stored data to an iPhone, indicate state using a tiny red/blue light, provide haptic feedback, detect motion/wear state, and support secure reset and factory programming.

Requirement	Frozen target
Finished exterior	51.0 x 21.0 x 11.0 mm maximum
Shape	Rounded capsule with proportions similar to PLAUD NotePin S; no copied private CAD
Mass	Target under 20 g

Requirement	Frozen target
Centre structure	Polycarbonate
Front and rear faces	0.60 mm brushed 5052-H32 aluminum
Radio region	Polycarbonate only over antenna; no metal, battery, motor, magnet, chain hardware, or conductive coating
Camera	None
PCB	45.8 x 18.0 mm current outline; 4-layer; 0.80 mm production target

4. Current electrical architecture

The ROB design uses:

- Raytac MDBT50Q-1MV2 nRF52840 radio/processor module.
- Nordic nPM1300 charger and power-management IC.
- Two Same Sky PDM microphones.
- Macronix 4-Gbit SLC serial NAND.
- ST LIS2DW12 accelerometer.
- USB-C with data, CC, and VBUS protection.
- Red/blue status LED, user button, haptic driver, factory pogo pads, battery lands, and motor lands.

The battery is a placeholder only. The required pack must be protected, three-wire, include a 10 kohm NTC, and fit within a maximum installed envelope of 26.0 x 15.5 x 4.8 mm including protection, insulation, folded tabs, tolerance, and swelling allowance.

5. Honest engineering status

The automated ROB verifier reports:

- 4 copper layers.
- 45.8 x 18.0 mm geometric outline.
- 70 electrical references.
- 223 source-of-truth pins checked.
- 0 geometric DRC violations.
- 176 unrouted PCB connections.**
- 0.50 mm provisional battery-to-antenna gap.
- 2.00 mm conservative motor-to-antenna gap.

Therefore, this is a strong starting package, not a factory release.

6. Who does what

Senior PCB engineer

The senior PCB engineer owns the electrical design until a release package exists. They must review every symbol and footprint, reconcile the 0.80/0.82 mm thickness issue, select or approve the exact battery, manually complete routing, review the nPM1300 loops against Nordic's reference, obtain the fabricator's impedance stack-up, run final ERC/DRC/net-equivalence checks, create a PCBA STEP, generate manufacturing outputs, and support first-board bring-up.

PCBA factory

The factory supplies or fabricates the bare PCB, sources approved SMT parts, prints solder paste, places components, reflows, inspects, programs/tests if quoted, and returns assembled PCBAs. The factory should build exactly two first articles and hold the other ten until written release.

Mechanical CAD engineer

The mechanical engineer designs the polycarbonate centre, aluminum faces, button actuator, light pipe, microphone acoustic wells, USB opening, battery retention, motor retention, chain/bail interface, sealing, and assembly strategy around the released PCBA STEP and exact battery STEP.

Final integration technician

The final technician receives tested PCBAs, batteries, motors, enclosure parts, firmware, assembly drawings, adhesives, insulation, acoustic mesh, and QA forms. They install the rear hand-solder parts, insulate and strain-relieve them, close the enclosure, cure adhesives, and run sealed-unit tests.

7. What each recipient receives

PCB engineer package

- Native KiCad `.kicad_pro`, `.kicad_sch`, `.kicad_pcb`, and `.kicad_dru`.
- Custom `.pretty` footprint library and `fp-lib-table`.
- Draft BOM, pin map, pad/net report, verifier, design generator, requirements, placement plan, stack-up request, and current DRC report.
- Nordic nPM1300 reference-layout archive.
- Exact scope of work and release checklist.

Mechanical CAD package

- Product envelope and material strategy.
- Provisional board-only STEP and board-outline DXF.
- PCBA placement image and interface requirements.
- Required CAD deliverable list.

The provisional files are for early packaging studies only. The final enclosure must use the engineer-released PCBA STEP and exact battery STEP.

PCBA factory package

Before release, the factory may receive only the RFQ and stack-up request. After release, it receives one locked ZIP containing Gerbers, drills, BOM, CPL, assembly drawings, fab drawing, stack-up, IPC-356, schematic PDF, PCBA STEP, programming instructions, firmware hash, and first-article test plan.

Final assembly package

- Released PCBA and battery STEP files.
- Complete enclosure STEP and exploded STEP.
- Polycarbonate production STEP/3MF.
- Aluminum front/rear/markings DXFs.
- Mechanical BOM.
- Adhesive and cure specification.
- Assembly traveler.
- Firmware and test instructions.
- Per-unit serial and QA record.

8. Materials and who supplies them

The pendant is a sandwich:

1. A 0.60 mm brushed 5052-H32 aluminum front face.
2. A polycarbonate centre frame that holds and electrically isolates the hardware.
3. The tested PCBA, protected battery, motor, insulation, foam, light pipe, and acoustic mesh.
4. A 0.60 mm brushed 5052-H32 aluminum rear face.
5. A polymer-only radio zone over the antenna.

You do not normally ship raw aluminum to the PCB engineer. The engineer specifies interfaces. The metal shop normally supplies certified aluminum sheet and cuts it from released DXFs. The PCBA factory supplies PCB laminate, copper, solder mask, finish, stencil, paste, and approved SMT parts unless the purchase order says Anticipy will consign a part. Anticipy should consign the exact approved battery and haptic motor for the first batch.

9. Seven-day rush attempt

The seven-day clock is a target, not a promise.

Day	Required outcome
Day 0	Hire PCB lead; send RFQs; freeze decision owner; select battery candidate; reserve

Day	Required outcome
	factory capacity
Day 1	Complete schematic/footprint review, routing, stack-up negotiation, mechanical interface review, and release-candidate checks
Day 2	Release manufacturing package only if every gate passes; factory starts two first articles; mechanical CAD continues around released PCBA
Day 3	Fabrication/assembly; firmware bench image and pogo fixture ready; enclosure prototypes printed
Day 4	Bring up FA-001 and FA-002; measure rails; program; test USB, BLE, microphones, storage, motion, LED, and haptic
Day 5	If both pass, release remaining ten PCBAs; freeze enclosure and cut/print parts
Day 6	Integrate and seal first two complete pendants; run acoustic/RF/charge/runtime tests
Day 7	Finish remaining units only if the first two sealed units pass; serialize, clean, and package

One to five working EVT units in a week may be possible with immediate engineering and a first-spin success. Ten polished sealed units in one week cannot be guaranteed.

10. Release sequence

The only safe sequence is:

1. Engineering release candidate.
2. Zero unconnected items and zero unresolved DRC errors.
3. Fabricator-reviewed 0.80 mm stack-up and 90-ohm USB geometry.
4. Exact battery approved.
5. Manufacturing package checksummed and signed.
6. Two first articles manufactured.
7. Both pass Gate A bench testing.
8. Written **RELEASE REMAINING TEN** approval.
9. Two sealed first articles pass Gate B.
10. Customer release.

11. Immediate owner actions

- Send the complete ZIP to one accountable senior PCB engineer.

- Send the factory RFQ immediately, but do not issue fabrication files or a purchase order.
- Make battery selection the first engineering decision.
- Assign one mechanical CAD owner and one firmware/bring-up owner.
- Reserve access to a current-limited bench supply, oscilloscope, DMM, SWD programmer, USB test host, BLE phone, audio stimulus, and pogo fixture.
- Use the project-control workbook to record one owner and deadline for every open gate.

12. What is deliberately missing

The following files cannot honestly be supplied yet because the design has not reached that stage:

- Released Gerbers and drills.
- Released BOM and CPL.
- Final PCBA STEP with all components.
- Exact battery STEP and approved battery MPN.
- Final enclosure STEP, production 3MF, and aluminum DXFs.
- Production firmware binary and SHA-256.
- Final programming fixture drawing and test software.

Their required filenames and acceptance rules are included so the responsible engineer knows exactly what must be produced.

APPENDIX A

Open items and decisions

P0 - blocks fabrication

Decision/item	Required evidence	Owner	Status
Exact battery	MPN, max drawing/STEP, protection, 10k NTC, polarity, limits, UN38.3, IEC 62133 evidence, MSDS, sample, stock	PCB lead + Anticipy	Open
Complete PCB routing	Zero unconnected items and reviewed routes	PCB lead	176 open connections
PMIC review	Schematic and layout signed against Nordic reference	PCB lead	Open
0.80 mm stack-up	Fabricator drawing, Dk, copper, dielectric, USB geometry	PCB lead + factory	Open
Thickness discrepancy	Explain/resolve 0.80 mm target versus 0.82 mm board-only STEP	PCB lead + mechanical	Open
Final PCBA STEP	All maximum-height components and interface bodies	PCB lead	Open
Factory package	Gerbers, drills, IPC-356, BOM, CPL, drawings, checksums	PCB lead	Open

P0 - blocks first-article bring-up

Decision/item	Required evidence	Owner	Status
Production/factory firmware	Reproducible source, binary, hash, flash/verify commands	Firmware lead	Open
NAND plan	ECC, bad blocks, filesystem, recovery and secure erase proven	Firmware lead	Open
Pogo fixture	Drawing, pin map, programmer and test connections	PCB + firmware	Open
Numeric test limits	Rail, ripple, current, audio, RF, thermal and runtime limits	PCB + firmware	Open

P1 - blocks sealed units

Decision/item	Required evidence	Owner	Status
Final enclosure	STEP, 3MF, DXFs, drawings, mechanical BOM	Mechanical lead	Open
Acoustic system	Mesh, port geometry, adhesive keepout, test limits	Mechanical + audio	Open
Antenna system	Polymer-only region and sealed-unit RF acceptance	PCB + mechanical	Open
Adhesive/seal process	Exact product, pattern, cure, rework and inspection	Mechanical + assembler	Open
Final integration owner	Named local technician, equipment, schedule and cost	Anticipy	Open

Decision log template

For every choice, record:

- Decision ID and date/time.
- Exact part/file/revision.
- Options considered.
- Evidence and test result.
- Decision and owner.
- Downstream files that must be updated.
- Whether a new revision/checksum is required.

APPENDIX B

Seven-day rush plan

Goal

Attempt to produce 12 custom-PCB EVT assemblies, with two first articles built and tested before the remaining ten. The practical success target is one to five working units in seven days. Ten sealed and polished units remain schedule-risky because revision-one PCB, battery, firmware, and enclosure work are still open.

Staffing

- One accountable senior PCB engineer.
- One firmware/bring-up engineer.
- One mechanical CAD engineer.
- One PCBA factory project owner.
- One final integration technician; two stations preferred after first-article release.
- One Anticipy decision owner who can approve or stop work immediately.

Schedule

Day	PCB/electrical	Mechanical	Firmware/test	Manufacturing/integration
0	Engineer accepts scope; review blockers	Start packaging study	Freeze bring-up plan	Send RFQs; reserve capacity
1	Battery decision; footprint review; complete routing; DRC/net checks	Develop centre/face concepts around provisional geometry	Create minimal bench image, self-tests, SWD plan	Factory returns stack-up, geometry, stock and schedule
2	Release only if all Gate 0 checks pass	Update from released PCBA STEP and battery	Release hashed FA firmware/test instructions	Start two first articles; print enclosure prototypes
3	Support DFM/CAM and answer exceptions	Fit/tolerance review	Prepare fixture, scripts, iPhone/BLE tools	Fabricate/assemble FA-001/FA-002
4	Lead current-limited bring-up	Correct mechanical issues found	Test rails, USB, BLE, audio, storage, motion, LED, haptic	Factory supplies inspection evidence
5	Close faults or issue remaining-ten release	Freeze enclosure after FA evidence	Freeze passed firmware hash	Assemble remaining ten only if both FAs pass
6	Support integration faults	Produce/finish parts	Sealed-unit regression	Integrate two sealed first articles
7	Final disposition	Final mechanical corrections	Overnight/runtime/backfill review	Integrate remaining passed units, serialize, clean, package

Daily decision gates

- End of Day 1: Can the board be released without waivers?
- End of Day 2: Did the factory accept the exact technology and files?
- End of Day 4: Did both open-board first articles pass?
- End of Day 6: Did both sealed first articles pass?

If any answer is no, preserve the hardware and continue the EVT correction loop. Do not convert an engineering failure into a pass to protect the calendar.

APPENDIX C

File dictionary

Files present now

File/type	Meaning	Primary user	Status
.kicad_pro	KiCad project settings	PCB engineer	Editable WIP
.kicad_sch	Electrical schematic	PCB engineer/firmware	Editable WIP
.kicad_pcb	Board outline, placement and future routing	PCB engineer	Editable WIP, unrouted
.kicad_dru	PCB design rules	PCB engineer	WIP
.pretty/ *.kicad_mod	Custom footprints	PCB engineer/assembler	Must be verified
BOM_DRAFT_DO _NOT_PRODUCE. csv	Draft part list	PCB engineer/procurement	Not procurement-released
pad_net_report.csv	Pin/net cross-check data	PCB engineer	Engineering reference
verify_project_r0b.py	Structural/release verifier	PCB engineer	Reports 176 unrouted items
placement.png/.svg	Visual component placement	All engineering roles	Provisional
PROVISIONAL_BOARD_ONLY.step	3D bare-board solid only	Mechanical engineer	Provisional, 0.82 mm export
PROVISIONAL_BOARD_OUTLINE.dxf	2D board outline	Mechanical engineer	Provisional

Files that must be created before fabrication

Required file	What it does	Creator
Release README with checksums	Locks revision and file set	PCB engineer
Gerber ZIP	Defines every manufactured PCB layer	PCB engineer
PTH/NPTH drill ZIP	Defines holes and slots	PCB engineer
Fabrication drawing	Defines material, finish, tolerances,	PCB engineer

Required file	What it does	Creator
	impedance and notes	
Approved stack-up	Defines real four-layer construction and USB geometry	Factory + PCB engineer
Released BOM	Controls procurement and substitutions	PCB engineer/procurement
CPL CSV	Controls placement X/Y/rotation/side	PCB engineer
Assembly drawings	Controls orientation and special process	PCB engineer
IPC-356	Electrical netlist for CAM/test	PCB engineer
PCBA STEP	Enables component collision review	PCB engineer
Programming/test package	Controls firmware and factory test	Firmware + PCB engineer

Files that must be created before enclosure/tooling release

Required file	What it does	Creator
Exact battery STEP/drawing	Controls battery fit and safety	Battery supplier + engineer
Enclosure assembly/exploded STEP	Controls complete mechanical fit	Mechanical engineer
Polycarbonate centre STEP/3MF	Controls structural centre	Mechanical engineer
Aluminum front/rear DXF	Controls sheet cutting	Mechanical engineer
Marking DXF	Controls ANTICIPY mark	Mechanical engineer
Mechanical drawing/BOM	Controls materials, tolerances and parts	Mechanical engineer
Assembly traveler	Controls sequence, adhesive, cure and inspection	Mechanical + final assembler

What to send when

- Send the entire ZIP now to the accountable PCB engineer.
- Send the master guide, mechanical folder and placement files now to the mechanical engineer.
- Send RFQs now to factories for capacity review only.

- Send final manufacturing outputs only after Gate 0 is signed.
- Send final assembly outputs only after both open-board first articles pass.

APPENDIX D

PCB engineer scope

Assignment

Take **ANT-PROD-ROB** from engineering work-in-progress to an EVT fabrication release for 12 PCBAs, with a mandatory hold after two first articles.

Starting point

- Editable KiCad schematic and PCB.
- Custom footprint library.
- 45.8 x 18.0 mm outline.
- Four-layer 0.80 mm target stack.
- 70 references and 223 checked design pins.
- 176 unrouted PCB connections.
- Draft BOM, pin map, requirements, placement plan, and Nordic reference archive.

Required work

Electrical review

- Review every schematic symbol, pin number, power pin, no-connect, polarity, and net label against the current manufacturer datasheet.
- Verify every footprint land pattern, exposed pad, paste aperture, maximum height, microphone port, USB shell slot, LED orientation, and module antenna keepout.
- Confirm the nPM1300 configuration, NTC circuit, buck startup values, charge limits, high-current grounds, and net ties against Nordic's reference.
- Confirm USB-C CC behavior, ESD topology, VBUSOUT connection, 27-ohm series resistors, and D+/D- polarity.
- Confirm the raw NAND choice is acceptable for the firmware schedule. If not, propose a lower-risk soldered storage part with a documented size, power, and firmware tradeoff.

Mechanical and sourcing decisions

- Select or approve the exact protected three-wire battery before release.
- Obtain its maximum-tolerance STEP/drawing, protection details, NTC curve, polarity, voltage/current/temperature limits, UN38.3, IEC 62133 evidence, MSDS, lot traceability, and stock confirmation.
- Reconcile the KiCad 0.80 mm target with the board-only STEP export that measures 0.82 mm.
- Review the battery, motor, USB, button, LED, microphone, pogo, and antenna interfaces with the mechanical engineer.

Layout completion

- Preserve an uninterrupted L2 ground reference.
- Keep all copper, components, and vias out of the Raytac all-layer antenna keepout.

- Route and review every connection manually; do not accept an autorouter result without net-by-net review.
- Keep nPM1300 switch loops and bypass paths compact and faithful to Nordic guidance.
- Route USB D+/D- as a controlled 90-ohm differential pair using the fabricator-approved stack-up and geometry.
- Review return paths, stitching, current density, via sizes, thermal reliefs, exposed pads, acoustic clearance, ESD placement, and test access.
- End with zero unconnected items and zero unresolved DRC errors.

Release outputs

Produce:

1. Complete editable KiCad project.
2. Final released schematic and PCB.
3. Footprint and symbol verification record.
4. ERC, DRC, net-equivalence, and unconnected-item reports.
5. Gerber X2/RS-274X ZIP.
6. Separate PTH and NPTH Excellon drill ZIP plus drill map.
7. IPC-356 netlist.
8. Released BOM with approved alternates and DNI flags.
9. CPL/pick-and-place file with origin and rotation convention.
10. Fabrication drawing and approved stack-up drawing.
11. Top and rear assembly drawings.
12. Full PCBA STEP model with maximum-height bodies.
13. SWD/programming/test-pad drawing.
14. Manufacturing README with SHA-256 checksums.
15. Bring-up support for FA-001 and FA-002.

Acceptance gates

No release until:

- Battery and enclosure interfaces are physically reviewed.
- The fabricator approves the exact 0.80 mm stack, minimum drill/pad, USB slots, controlled-impedance geometry, and rush schedule in writing.
- Every part is available, approved, or explicitly consigned.
- Zero unrouted connections remain.
- Zero unresolved ERC/DRC/net-equivalence issues remain.

- PMIC, USB, RF, acoustic, thermal, and manufacturing reviews are signed.
- Output files are regenerated from the final native source and checksummed together.

Required response from candidate engineer

Confirm:

- Relevant nRF52/nPM1300, USB, RF module, compact wearable, and KiCad experience.
- Who personally performs layout and review.
- Start time and committed 24-36 hour engineering sprint availability.
- Availability for first-board bring-up.
- Fixed or capped estimate and what would stop the sprint.

APPENDIX E

Mechanical CAD handoff

Objective

Create a manufacturable rounded-capsule enclosure no larger than 51.0 x 21.0 x 11.0 mm around the released Anticipy PCBA, exact protected battery, 10 mm haptic motor, light pipe, microphone paths, USB-C, button, and chain/bail interface.

Inputs available now

- Provisional 45.8 x 18.0 mm board-only STEP.
- Provisional board-outline DXF.
- Placement PNG/SVG.
- Electrical requirements and antenna keepout rules.
- Provisional battery and motor envelopes in KiCad.

These are early packaging inputs only. The current STEP exports at 0.82 mm and contains no component bodies. Do not release tooling, aluminum, or final fit from it.

Inputs required before final release

- Engineer-released PCBA STEP with maximum-height component bodies.
- Exact battery STEP and controlled maximum-tolerance drawing.
- Motor drawing and wire exit.
- Button travel/force and access target.
- USB connector insertion envelope and tolerance.
- LED emitting centre and light-pipe optical requirements.
- Microphone port centres, supplier keepout, mesh and acoustic targets.
- Exact Raytac antenna keepout and polymer-only external zone.

Material architecture

- Front: 0.60 mm brushed 5052-H32 aluminum.
- Centre: injection-grade polycarbonate geometry; EVT may be printed in polycarbonate if tolerances are verified.
- Rear: 0.60 mm brushed 5052-H32 aluminum.
- Antenna end: polycarbonate only.
- Internal: flame-retardant electrical insulation, battery liner, anti-rattle foam, acoustic mesh, light pipe, controlled adhesive, and strain relief.

Required CAD outputs

1. ANT-PROD-R0B_ENCLOSURE_ASSEMBLY.step
2. ANT-PROD-R0B_EXPLODED_ASSEMBLY.step

3. ANT-PROD-R0B_POLYCARBONATE_CENTRE.step
4. ANT-PROD-R0B_POLYCARBONATE_CENTRE.3mf
5. ANT-PROD-R0B_FRONT_ALUMINUM.dxf
6. ANT-PROD-R0B_REAR_ALUMINUM.dxf
7. ANT-PROD-R0B_ANTICIPY_MARKING.dxf
8. ANT-PROD-R0B_LIGHT_PIPE.step
9. ANT-PROD-R0B_BUTTON_ACTUATOR.step
10. ANT-PROD-R0B_MECHANICAL_DRAWING.pdf
11. ANT-PROD-R0B_EXPLODED_ASSEMBLY.pdf
12. ANT-PROD-R0B_MECHANICAL_BOM.csv
13. ANT-PROD-R0B_TOLERANCE_AND_ADHESIVE_SPEC.pdf

Design checks

- No aluminum, battery, motor, magnet, bail, chain hardware, or conductive coating enters the antenna zone.
- Battery cannot bend, slide, rattle, touch exposed copper, or move toward the antenna after swelling and adhesive creep.
- USB plug reaches full insertion in both orientations without levering the PCB.
- Button is repeatable and cannot remain pressed after a drop.
- Red/blue LED aligns with the light pipe without leaking into the microphones.
- Microphone ports remain clean, short, symmetric, sealed, and protected by acoustic mesh.
- Motor is mechanically isolated enough to avoid unacceptable microphone noise.
- Enclosure can be assembled and reopened during EVT without destroying the PCBA.
- Both sealed first articles pass RF, acoustic, charge, thermal, drop, rattle, and runtime tests.

APPENDIX F

Firmware handoff

Status

No production firmware binary is included. This document defines the firmware work needed for ROB bring-up and release.

Platform

- MCU/radio: nRF52840 in Raytac MDBT50Q-1MV2.
- Power/charger: Nordic nPM1300 over I2C.
- Audio: two PDM microphones on shared clock with separate data source paths.
- Storage: Macronix MX35LF4GE4AD-Z4I 4-Gbit SLC serial NAND.
- Motion: LIS2DW12 over I2C with two interrupts.
- USB: nRF52840 USB 2.0 device through protected D+/D-.
- BLE: secure control and live/stored audio transfer.

Required behaviors

- Boot safely at the ROB power defaults; raise BUCK1 from 2.7 V to 3.0 V only after the nPM1300 is identified and configured.
- Configure charging only after the exact battery profile and NTC are approved.
- Record each microphone channel, verify channel assignment, and support ambient recording without a button press.
- Implement audio framing, timestamps, corruption recovery, and backfill to iPhone.
- Implement raw NAND ECC, bad-block discovery/management, wear strategy, power-fail recovery, filesystem, secure erase, and near-capacity behavior.
- Support authenticated BLE provisioning, transfer, reset, and firmware update.
- Support USB data and recovery/programming diagnostics.
- Read motion/orientation and motion interrupts.
- Drive red/blue status patterns through nPM1300 LED sinks.
- Drive haptic patterns through the MOSFET and avoid audio corruption/resets.
- Read button events through the nPM1300 interrupt path, not a direct MCU GPIO.
- Expose factory self-tests and measurement logs.

Required firmware deliverables

1. Source repository and reproducible build instructions.
2. Locked toolchain/SDK versions.
3. Pin-map header matching [ROB_PINMAP.md](#).
4. Factory-test image and production image.
5. [.hex](#) or [.bin](#) plus SHA-256 hashes.

6. SWD flash/verify command.
7. Serial-number and provisioning method.
8. Factory test command list and machine-readable pass record.
9. Power-mode table with measured current limits.
10. BLE service/protocol specification.
11. Storage format/recovery specification.
12. Secure reset and firmware-update specification.
13. Known-issues and rollback procedure.

Bring-up order

1. Power rails and PMIC identity.
2. SWD erase/program/verify and UART logging.
3. I2C scan, nPM1300, and accelerometer.
4. USB enumeration in both cable orientations.
5. BLE advertising, connect, security, and transfer.
6. Microphone capture and channel validation.
7. NAND identity, ECC, bad-block scan, write/read/hash.
8. LED, button, and haptic.
9. Approved battery/NTC and charging.
10. Sealed-unit audio, RF, storage, runtime, and power-fail testing.

APPENDIX G

PCBA factory handoff

Status

This folder contains an RFQ and capability package only. It is not a fabrication release.

Planned build

- 12 four-layer PCBAs.
- Approximately 45.8 x 18.0 x 0.80 mm.
- Components on top side only for one SMT reflow pass.
- Rear battery and haptic motor installed by hand only after programming and bare-PCBA test.
- Build exactly two first articles, then hold the other ten pending written release.

Files the factory will receive after engineering release

- Release README and checksums.
- Gerber ZIP.
- PTH/NPTH drill ZIP and drill map.
- Fabrication drawing and approved stack-up.
- Released BOM with sourcing and substitution controls.
- CPL/pick-and-place CSV.
- Top and rear assembly drawings.
- Schematic PDF and IPC-356 netlist.
- PCBA STEP.
- Programming/test instructions and firmware hash.
- First-article test plan.

Required factory confirmations

- 0.80 mm four-layer high-Tg FR-4 is physically in stock.
- 5/5 mil, 0.20/0.45 mm standard through-vias, and two 0.16/0.36 mm USB fanout vias are accepted.
- GCT USB4500 mid-mount slots are accepted.
- Exact 90-ohm USB geometry and panel coupon/TDR will be provided.
- 0201, QFN exposed pad, WSON exposed pad, one top-side stencil, and one reflow pass are supported.
- AOI and X-ray of exposed-pad devices are available.
- No substitution or CAM edit occurs without written approval.
- Two first articles can be assembled and the remaining ten held.

APPENDIX H

Final assembly SOP

Status: planning document only. Final adhesive, torque, cure, acoustic, and test values must come from the released mechanical package.

Inputs per unit

- One passed and serialized PCBA.
- One approved protected three-wire battery.
- One tested haptic motor.
- One polycarbonate centre.
- One front and one rear aluminum face.
- Light pipe and button actuator.
- Acoustic mesh for both microphones.
- Electrical insulation/battery liner.
- Anti-rattle foam and strain relief.
- Released adhesives and cleaning materials.
- Bail/chain hardware and packaging.

Assembly order

1. Verify the serial number, PCBA test record, firmware hash, battery lot, motor lot, and enclosure revision.
2. Inspect microphones, USB connector, antenna region, pogo pads, and solder joints. Quarantine contamination or damage.
3. Program and electrically test through the rear pogo fixture while BT1 and M1 are absent.
4. Measure battery polarity and NTC. Hand-solder BT1 in the frozen order: pin 1 VBAT, pin 2 NTC, pin 3 GND.
5. Add released insulation and strain relief. Confirm the battery cannot touch exposed copper or move toward the antenna.
6. Hand-solder M1 with documented polarity. Add strain relief and mechanical isolation.
7. Retest power, USB, BLE, microphones, storage, motion, LED, button, and haptic before closing.
8. Install the PCBA/battery/motor subassembly into the polycarbonate centre without forcing or bending.
9. Install acoustic mesh, light pipe, button actuator, foam, and seals. Keep adhesive and debris away from microphone ports, USB, and antenna zone.
10. Attach front and rear aluminum faces using the released adhesive pattern and cure process.
11. Run shake/rattle, visual-gap, USB insertion, button, light-pipe, microphone, BLE, charge, thermal, storage/backfill, and runtime tests.
12. Record every result against the serial number. Clean and package only passed units.

Stop conditions

Stop for smoke, odor, swelling, heat, unstable current, incorrect polarity, damaged microphone, short, programming failure, reset, data corruption, RF loss, audio blockage, rattle, adhesive intrusion, or any unrecorded deviation.

APPENDIX I

Release and QA gates

Gate 0 - engineering release

- [] Exact battery MPN, drawing, STEP, protection, NTC, polarity, compliance evidence, sample, and stock approved.
- [] Every symbol, pin map, footprint, polarity, height, and acoustic port reviewed.
- [] PMIC loops and grounds reviewed against Nordic reference.
- [] Fabricator returns and Anticipy approves the 0.80 mm stack-up and 90-ohm USB geometry.
- [] 0.80/0.82 mm thickness discrepancy closed.
- [] Zero unconnected items.
- [] Zero unresolved ERC, DRC, or net-equivalence issues.
- [] Final PCBA STEP passes enclosure/battery/motor/USB/button/LED/microphone/antenna collision review.
- [] Released BOM, CPL, Gerbers, drills, IPC-356, fab/assembly drawings, STEP, test files, and checksums agree.

Gate A - release remaining ten PCBAs

Both FA-001 and FA-002 must pass:

- [] Revision, checksum, material, dimensions, electrical test, AOI, and X-ray evidence.
- [] Unpowered short/continuity/isolation checks.
- [] Current-limited power-up and all rail measurements.
- [] SWD program/verify, device ID, serial, reset, UART, I2C, PMIC interrupt, and button.
- [] USB power/data in both orientations.
- [] BLE advertising, security, control, and transfer.
- [] Both microphone channels, quiet/tone/speech, BLE coexistence, and haptic-noise test.
- [] NAND identity, ECC, bad blocks, write/read/hash, power-fail recovery, secure erase, and near-capacity behavior.
- [] LED, haptic, accelerometer, and approved battery/charge tests.
- [] No open deviation or missing measurement.

Gate B - sealed-unit release

Both sealed first articles must pass:

- [] Fit and retention with no collision, rattle, chafe, or exposed battery risk.
- [] Microphone acoustic response and contamination inspection.
- [] USB insertion, button travel, and light-pipe alignment.
- [] Sealed-device BLE range/RSSI/packet-loss limits.
- [] Charge, thermal, drop, shake, overnight recording, backfill, secure reset, and runtime tests.
- [] Complete traceability: PCB lot, battery lot, motor lot, firmware hash, enclosure revision, serial, and test record.

Approval rule

Verbal approval is not release. The approval must identify the exact revision, file checksums, two passing first-article serials, known deviations, and approver.

APPENDIX J

Supplier contacts

Verified on 2026-08-29 from official company pages.

Organization	Use	Contact
CCI Canadian Circuits, Surrey	Primary rush PCB/PCBA capability request	sales@canadiancircuits.com , 604-599-8600, https://www.canadiancircuits.com/contact/
Enigma Interconnect, Burnaby	Local bare-board backup	boards@enigmacorp.com , 604-420-3313, https://www.enigmacorp.com/quote
MistyWest, Vancouver	Full-service hardware engineering option	contact@mistywest.com , 604-292-7036, https://www.mistywest.com/contact/

These contacts are leads, not booked commitments. Ask for written confirmation of scope, owner, start time, price, material stock, and exact delivery date.